

Spectral Analysis Report

Molecule: Ethyl Benzoate (Test v1.4)

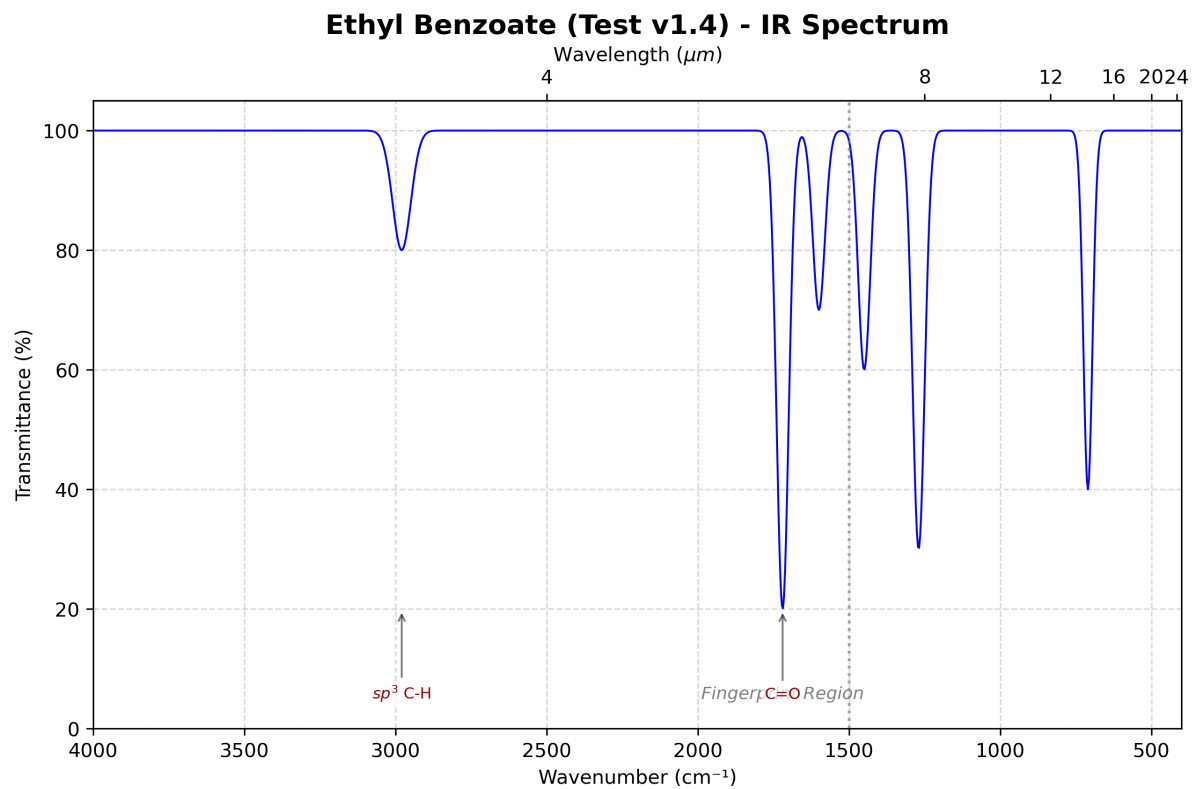
IUPAC Name: Ethyl Benzoate

Common Name: Ethyl Benzoate

Molecular Formula: $C_9H_{10}O_2$

Date: 2026-03-03 23:20:45

Infrared Spectrum (IR)

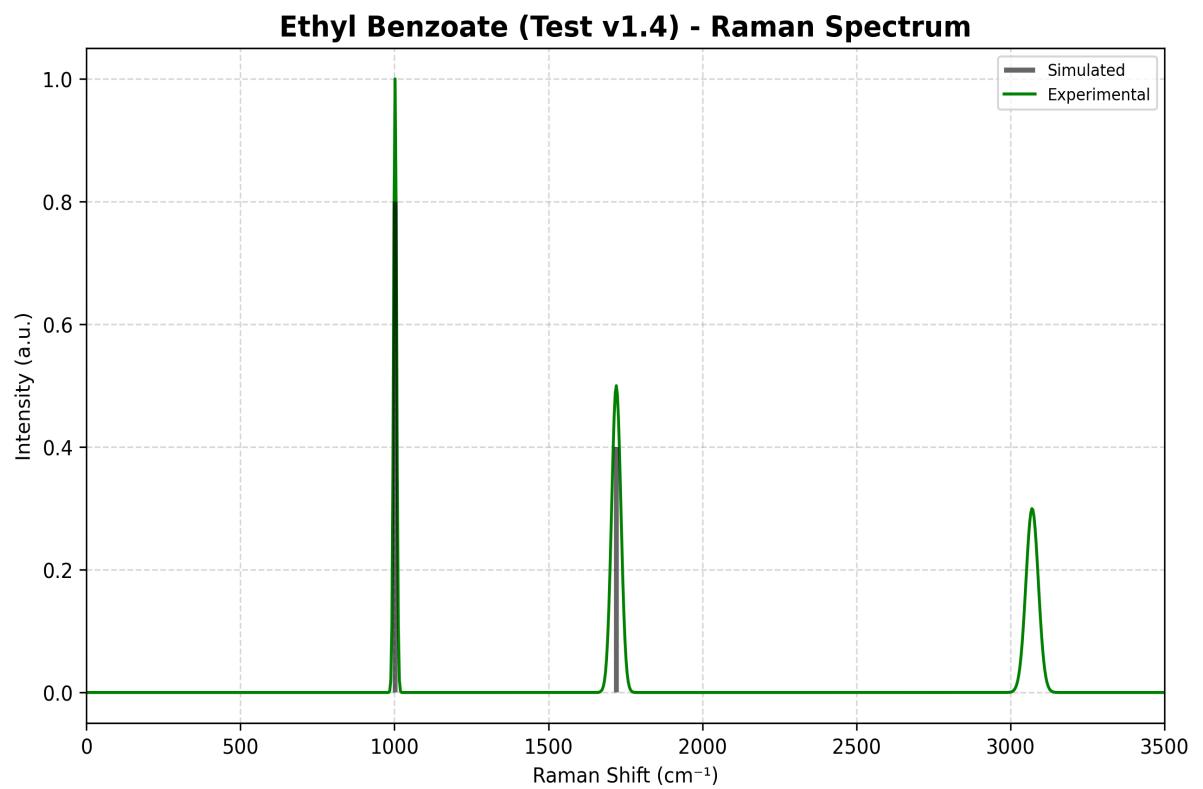


Peak List:

Frequency / Shift	Intensity
2980.00	20.00
1720.00	20.00
1270.00	30.00
710.00	40.00

Interpretation: Strong C=O stretch at 1720 cm⁻¹, C-O stretch at 1270 cm⁻¹.

Raman Spectrum



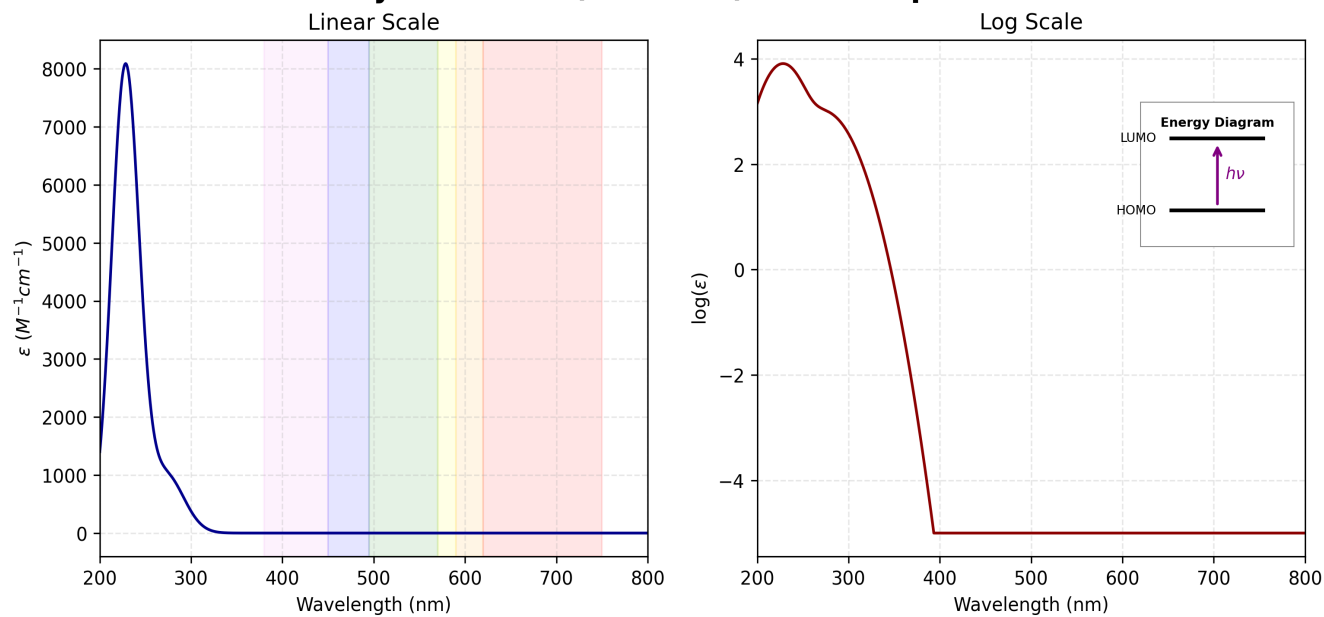
Peak List:

Frequency / Shift	Intensity
1002.00	1.00
1720.00	0.50

Interpretation: Ring breathing mode characteristic of mono-substituted benzene.

UV-Vis Spectrum

Ethyl Benzoate (Test v1.4) - UV-Vis Spectrum

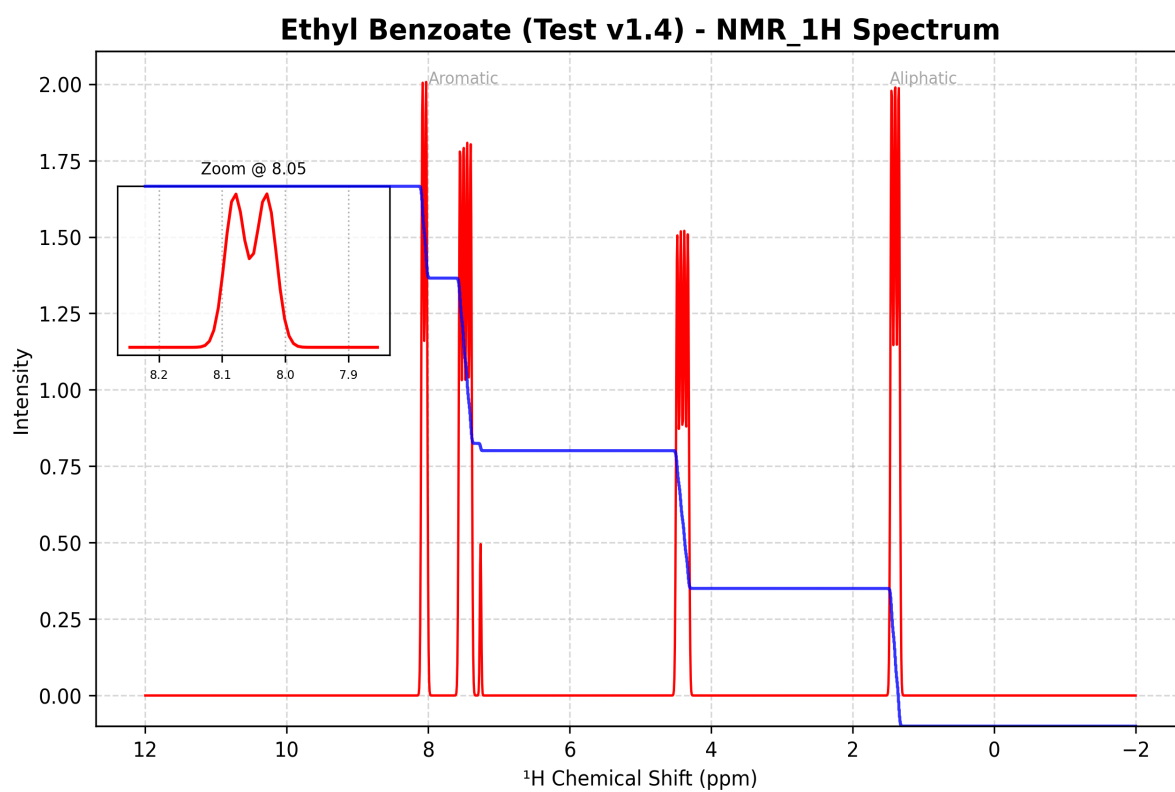


Peak List:

Frequency / Shift	Intensity
228.00	0.80
272.00	0.10

Interpretation: Pi-Pi* transition at 228 nm.

^1H NMR Spectrum

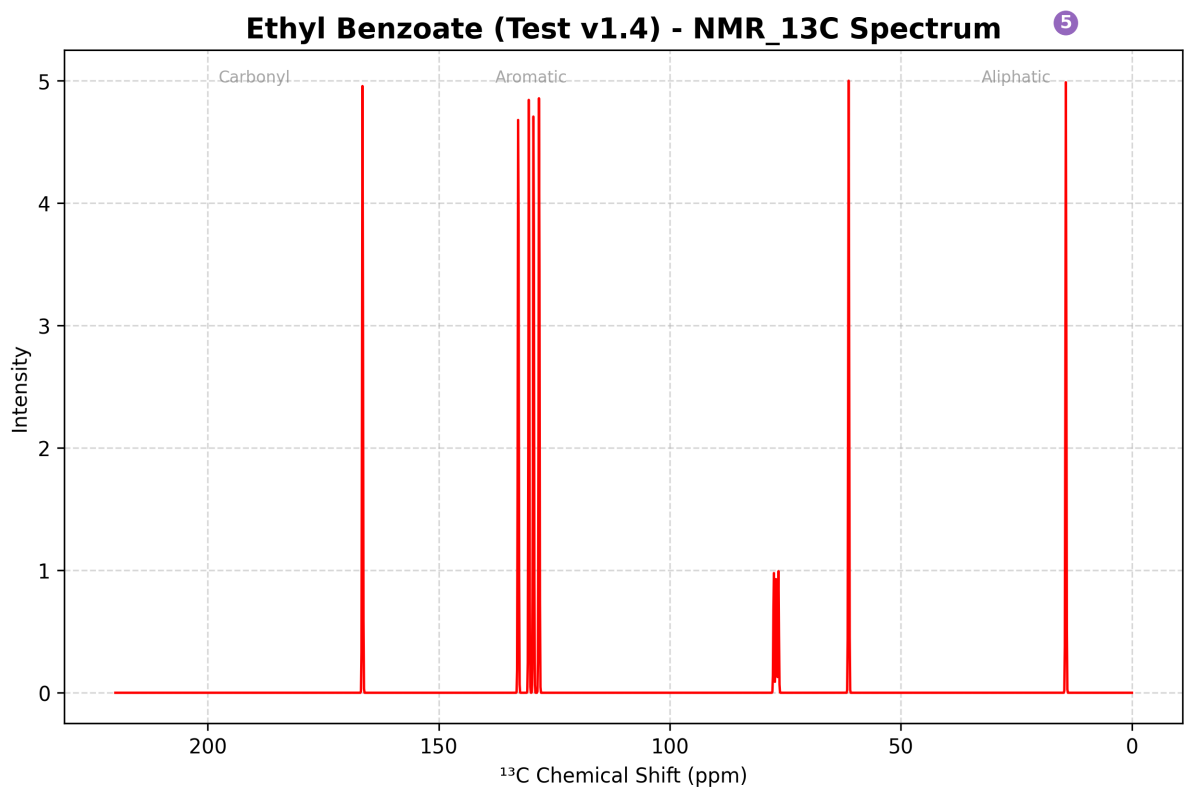


Peak List:

Frequency / Shift	Intensity
8.05	10.00
7.50	8.00
7.40	8.00
4.40	6.00
1.40	8.00

Interpretation: Ethyl group (q, t) and aromatic region (m).

¹³C NMR Spectrum



Peak List:

Frequency / Shift	Intensity
166.50	10.00
132.80	8.00
128.30	8.00
61.30	6.00
14.30	5.00

Interpretation: Carbonyl at 166.5 ppm, O-CH₂ at 61.3 ppm.